

Analysis of p-adic Dynamical Systems for Some Rational Functions in the Projective Line $\mathbb{P}^1(\mathbb{Q}_p)$

Toka Diagana

The University of Alabama in Huntsville (USA)

e-mail: toka.diagana@uah.edu

Wuhan University
Wuhan, CHINA

OUTLINE

1 SUMMARY

2 NON-ARCHIMEDEAN VALUED FIELDS

- BASIC DEFINITIONS
- P-ADIC NUMBERS

3 THE PROJECTIVE LINE

4 P-ADIC DYNAMICAL SYSTEMS

5 THE LONG-TERM BEHAVIOR OF SBH P-ADIC DS

- THE POLES OF f_N
- FIXED-POINTS OF f_N
- THE B-H P-ADIC DS
 - CASE $a = 1$
 - CASE $a \neq 1$
- THE CASE OF SBH P-ADIC DS

6 OPEN QUESTION

7 REFERENCES

- This talk is devoted to the analysis of the **(discrete) p-adic dynamical systems associated with the Sigmoid Beverton-Holt model in the projective line $\mathbb{P}^1(\mathbb{Q}_p)$.**
- The **Sigmoid Beverton-Holt (SBH) model** is a **discrete population model** that has a wide range of practical applications. For instance, it is utilized to:
 - **predict insurance premiums**
 - **forecast future changes in populations**, etc.
- The **fixed points, maximal Siegel disks, attractors, periodic trajectories, and the Julia and Fatou sets** associated with the above p-adic dynamical systems are analyzed.
- Some **open questions** will also be outlined at the end of the talk.

Definition

An **absolute value** on a field $(\mathbb{K}, +, \cdot)$ is a mapping $|\cdot| : \mathbb{K} \rightarrow [0, \infty)$ that satisfies, for all $x, y \in \mathbb{K}$,

① $|x| = 0$ iff $x = 0$;

② $|xy| = |x||y|$; and

③ $|x + y| \leq |x| + |y|$.

If (3) is replaced with the inequality:

④ $|x + y| \leq \max(|x|, |y|)$, with equality if, $|x| \neq |y|$; one says that $|\cdot|$ is a **non-archimedean absolute value**.

- Obviously, (4) yields (3).
- The inequality in (4) is called the **non-archimedean triangle inequality**.
- The (non-archimedean) absolute value $|\cdot|$ defined above induces a (non-archimedean) metric d on $\mathbb{K} \times \mathbb{K}$ defined by
$$d(x, y) = |x - y| \quad \forall x, y \in \mathbb{K}.$$

- A non-archimedean valued field $(\mathbb{K}, |\cdot|)$ is said to be **complete** if the metric space $(\mathbb{K}, d)$ is complete.

Classical examples of non-archimedean valued fields include for instance $(\mathbb{Q}_p, |\cdot|_p)$ and $(\mathbb{C}_p, |\cdot|_p)$.

Let $p \geq 2$ be a prime. If $n \in \mathbb{Z} \setminus \{0\}$, then the p -adic valuation of n is defined by $v_p(n) = \max\{t : p^t | n\}$. If $a \in \mathbb{Q} \setminus \{0\}$ and $a = \frac{m}{n}$, then we set

$$v_p(a) = v_p(m) - v_p(n).$$

We also set $v_p(0) = \infty$. The mapping $v_p : \mathbb{Q} \rightarrow \mathbb{Z} \cup \{\infty\}$, is a valuation on $\mathbb{Q}$, that is,

- a) $v_p(a) = \infty$ if and only if $a = 0$;
- b) $v_p(ab) = v_p(a) + v_p(b)$; and
- c) $v_p(a + b) \geq \min(v_p(a), v_p(b))$ with equality if $v_p(a) \neq v_p(b)$

for all $a, b \in \mathbb{Q}$.

Definition

For every $x \in \mathbb{Q}$, define the p -adic absolute value of x as follows,

$$|x|_p = \begin{cases} p^{-v_p(x)} & \text{if } x \neq 0, \\ 0 & \text{if } x = 0. \end{cases}$$

Consider the following examples:

Example

① Set $p = 3$. Then, $|6|_3 = |2^1 \times 3^1|_3 = 3^{-1} = \frac{1}{3}$.

② Set $p = 2$. Then, $\left|\frac{1}{8}\right|_2 = \frac{|1|_2}{|8|_2} = \frac{|2^0|_2}{|2^3|_2} = \frac{2^{-0}}{2^{-3}} = \frac{1}{2^{-3}} = 8$.

The absolute value $|\cdot|_p : \mathbb{Q} \mapsto \mathbb{R}_+$ is a *non-archimedean* absolute value on $\mathbb{Q}$. This is, $|\cdot|_p$ satisfies the following conditions,

- a) $|a|_p = 0$ if and only if $a = 0$;
- b) $|ab|_p = |a|_p \cdot |b|_p$; and
- c) $|a + b|_p \leq \max(|a|_p, |b|_p)$ with equality if $|a|_p \neq |b|_p$.

for all $a, b \in \mathbb{Q}$.

The metric space $(\mathbb{Q}, d_p)$, where $d_p(a, b) = |a - b|_p$ for all $a, b \in \mathbb{Q}$, is not **topologically complete**. Its completion denoted $(\mathbb{Q}_p, d_p)$, which is also a field, is called the **field of p-adic numbers**.

We have the following characterization of p -adic numbers:

Lemma

- a) *Every element $x \in \mathbb{Q}_p$ can uniquely be expressed as (Hensel's expansion)*

$$x = \sum_{t \geq -t_0} a_t p^t \text{ where } 0 \leq a_t \leq p-1.$$

- b) *The ring of p -adic integers,*

$\mathbb{Z}_p = \{x \in \mathbb{Q}_p : |x|_p \leq 1\} = \{x \in \mathbb{Q}_p : v_p(x) \geq 0\}$, *can be rewritten as follows:*

$$\mathbb{Z}_p = \left\{ x \in \mathbb{Q}_p : x = \sum_{t=0}^{\infty} a_t p^t, 0 \leq a_t \leq p-1 \right\}.$$

- c) $\mathbb{Z}$ is dense in $\mathbb{Z}_p$.
 d) *The maximal ideal of $\mathbb{Z}_p$ is $p\mathbb{Z}_p$.*

Corollary

The ring of p -adic integers $\mathbb{Z}_p$ is compact. Furthermore, the residue field of $\mathbb{Z}_p$ is the finite field, $\mathbb{F}_p = \mathbb{Z}_p/p\mathbb{Z}_p$.

In order to locate the zeros of some p -adic polynomials, one often makes use of the so-called **Hensel Lemma**.

In this talk, we will use a variant of the Hensel Lemma known as the **Hensel Lifting Lemma**.

This Lemma states, in essence, that **if a polynomial with p -adic integer coefficients whose reduction map modulo p has a root in the finite field $\mathbb{F}_p = \mathbb{Z}/p\mathbb{Z}$, then such a solution can be lifted to the ring of p -adic integers $\mathbb{Z}_p$.**

In order to simplify the notations, if $g \in \mathbb{Z}_p[X]$; **its reduction map modulo p , that is, $\bar{g} : \mathbb{F}_p \mapsto \mathbb{F}_p$ (respectively, $\bar{g}' : \mathbb{F}_p \mapsto \mathbb{F}_p$) will simply be denoted g (respectively, g').**

Lemma

(Hensel Lifting Lemma) *If $F \in \mathbb{Z}_p[X]$ and $\alpha \in \mathbb{F}_p = \mathbb{Z}/p\mathbb{Z}$ such that $F(\alpha) = 0$ in $\mathbb{F}_p$, $F'(\alpha) \neq 0$ in $\mathbb{F}_p$, then there is a unique $\beta \in \mathbb{Z}_p$ such that $F(\beta) = 0$.*

Let $P \in \mathbb{Q}_p[X]$. Then $P(X) = 0$ may or may not have solutions in $\mathbb{Q}_p$. Therefore, $\mathbb{Q}_p$ **is not algebraically closed**.

Example

Let $p \geq 2$ be a prime and consider $P(X) = X^2 - p \in \mathbb{Q}_p[X]$. If $\omega \in \mathbb{Q}_p$ were a root of P , then we would have $\omega^2 = p$ which yields $|\omega|_p^2 = \frac{1}{p}$.

Given $|\mathbb{Q}_p^*|_p = p^{\mathbb{Z}}$, then $|\omega|_p = p^{-\frac{1}{2}}$ would give a contradiction. In conclusion, P has no roots in $\mathbb{Q}_p$.

- In view of the above, $\mathbb{Q}_p$ is not algebraically closed.
- Let $\overline{\mathbb{Q}_p}^{alg}$ be its algebraic closure.
- It can be shown that $|\cdot|_p$ can be extended to $\overline{\mathbb{Q}_p}^{alg}$.

- Further, $|\cdot|_p$ on $\overline{\mathbb{Q}_p}^{alg}$ is a valuation extending $|\cdot|_p$ on $\mathbb{Q}_p$.
- The (topological) completion of $\overline{\mathbb{Q}_p}^{alg}$ under $|\cdot|_p$ is denoted $\mathbb{C}_p$ and is called the **field of complex p-adic numbers**.
- $|\mathbb{C}_p^*|_p = p^{\mathbb{Q}}$.
- Not only $\mathbb{C}_p$ **is complete, but it is also algebraically closed, and is a non-archimedean valued field which is excellent for analysis**, in the same standing as $\mathbb{C}$, the field of complex numbers.
- Note that $\mathbb{C}_p$ and $\mathbb{C}$ are **isomorphic as fields but as valued fields, they are quite different**.
- Moreover, $\mathbb{C}_p$ **is separable, of infinite dimension as a vector space over $\mathbb{Q}_p$, but it is not locally compact**.
- Moreover, the **residue field of $\mathbb{C}_p$ is the algebraic closure of the finite field $\mathbb{F}_p$** , which is given by

$$\overline{\mathbb{F}_p}^{alg} = \bigcup_{n \geq 1} \mathbb{F}_{p^n}.$$

Definition

Let $(\mathbb{K}, |\cdot|)$ be a non-archimedean valued field. **The projective line on $\mathbb{K}$ which is denoted by $\mathbb{P}^1(\mathbb{K})$, is the the set of lines in $\mathbb{K} \times \mathbb{K}$ that contains the origin.** The field $\mathbb{K}$ is embedded in $\mathbb{P}^1(\mathbb{K})$ through the following mapping $x \mapsto [x, 1]$. The point $[1, 0]$ is the point at infinity. There is a natural identification between $\mathbb{P}^1(\mathbb{K})$ and $\mathbb{K} \cup \{\infty\}$. If $P_1 = [x_1, y_1]$ and $P_2 = [x_2, y_2]$ are two points in the projective line $\mathbb{P}^1(\mathbb{K})$, their **chordal metric** is defined by

$$d_c(P_1, P_2) = \frac{|x_1 y_2 - x_2 y_1|}{\max(|x_1|, |y_1|) \cdot \max(|x_2|, |y_2|)}.$$

If P_1, P_2 are in $\mathbb{K}$, one can take $y_1 = y_2 = 1$, and identify P_1 with x_1 and P_2 with x_2 . In this case, the **chordal distance** between P_1 and P_2 is

$$d_c(x_1, x_2) = \frac{|x_1 - x_2|}{\max(|x_1|, 1) \cdot \max(|x_2|, 1)}.$$

Moreover,

$$d_c(x_1, \infty) := \lim_{x_2 \rightarrow \infty} d_c(x_1, x_2)$$

where $d_c(x_1, \infty) = |x_1|$ if $|x_1| \leq 1$ and $d_c(x_1, \infty) = \frac{1}{|x_1|}$ if $|x_1| > 1$. Obviously, $d_c(\infty, \infty) = 0$.

Proposition (J. Silverman's Book)

The chordal metric has the following properties: for all $P_1 = [x_1, y_1]$, $P_2 = [x_2, y_2]$ and $P_3 = [x_3, y_3]$ in $\mathbb{P}^1(\mathbb{K})$,

- a) $0 \leq d_c(P_1, P_2) \leq 1$;
- b) $d_c(P_1, P_2) = 0$ if and only if $P_1 = P_2$;
- c) $d_c(P_1, P_2) = d_c(P_2, P_1)$; and
- d) $d_c(P_1, P_3) \leq \max(d_c(P_1, P_2), d_c(P_2, P_3))$ with equality if $d_c(P_1, P_2) \neq d_c(P_2, P_3)$.

Setting $\infty^{-1} = 0$ and $0^{-1} = \infty$ it follows that $d_c(x_1^{-1}, x_2^{-1}) = d_c(x_1, x_2)$. We also have,

$$d_c(x_1, x_2) \leq |x_1 - x_2|$$

for all $x_1, x_2 \in \mathbb{K}$.

- If $\max(|x_1|, |x_2|) \leq r$, then

$$d_c(x_1, x_2) \geq \frac{|x_1 - x_2|}{(\max(1, r))^2}.$$

In particular, if $r \leq 1$, then $d_c(x_1, x_2) = |x_1 - x_2|$. Moreover, if $1 \leq \min(|x_1|, |x_2|)$, then

$$d_c(x_1, x_2) = |x_1^{-1} - x_2^{-1}|.$$

- Both $|\cdot|$ and d_c induce the same topology on $\mathbb{K}$. Moreover, if the non-archimedean valued field $(\mathbb{K}, |\cdot|)$ is complete, then so is $(\mathbb{P}^1(\mathbb{K}), d_c)$.
- If $\mathbb{K}$ is locally compact, then $\mathbb{P}^1(\mathbb{K})$ identified with $\mathbb{K} \cup \infty$, is the Alexandroff compactification of $\mathbb{K}$.

A (discrete) dynamical system is a pair $(\mathbb{F}, f)$ consisting of a set $\mathbb{F}$ and a mapping $f : \mathbb{F} \mapsto \mathbb{F}$.

For a given $x \in \mathbb{F}$, the **(forward) orbit** of x denoted $O_f(x)$ is the set of its iterations

$$O_f(x) = \{x, f(x), f^{[2]}(x), f^{[3]}(x), \dots, f^{[k]}(x), \dots\}$$

where $f^{[k]} = f \circ f \circ f \circ \dots \circ f$ and $f^{[0]} = I$ (identity).

A point $x_0 \in \mathbb{F}$ is said to be **periodic** if $f^{[k]}(x_0) = x_0$ for some integer $k \geq 1$. The smallest k satisfying this property is called the exact period of x_0 .

A point $x_0 \in \mathbb{F}$ such that $f(x_0) = x_0$ (equivalently, a 1-periodic point) is called a **fixed-point** of f . The set of fixed-points of f is denoted $Fix(f)$.

If $O_f(x_0)$ finite (equivalently, if some iterate $f^{[k]}(x_0)$ is periodic), then the point $x_0 \in \mathbb{F}$ is said to be **preperiodic**.

If $O_f(x_0)$ infinite, then the point $x_0 \in \mathbb{F}$ is said to be **wandering**.

Example

Suppose $\mathbb{F} = \mathbb{Z}$ and consider the mapping $f : \mathbb{Z} \mapsto \mathbb{Z}, z \mapsto z^2 - 2$.

-1 and 2 are the fixed points of f .

0 and -2 are preperiodic points of f .

$3, 4, 5$ are wandering points.

Suppose $(\mathbb{F}, d)$ is a metric space. The Fatou and Julia sets are respectively defined by:

Definition

A Fatou set $\mathcal{F}_f$ is defined as the set of all $x_0 \in \mathbb{F}$ for which the family of iterations $\{f^{[n]}\}_{n \geq 0}$ is equicontinuous on some neighborhood of x_0 .

and

Definition

A Julia set $\mathcal{J}_f$ is defined as the complement in $\mathbb{E}$ of the Fatou set. That is, $\mathcal{J}_f = \mathbb{F} \setminus \mathcal{F}_f$

Let $(\mathbb{K}, |\cdot|)$ be a complete non-archimedean valued field and let

$$(\mathbb{F}, d) = (\mathbb{P}^1(\mathbb{K}), d_c).$$

We need the following definition:

Definition

If x_0 is a periodic point of $f : \mathbb{P}^1(\mathbb{K}) \mapsto \mathbb{P}^1(\mathbb{K})$ of period m , then $\lambda_m := (f^{[m]})'(x_0)$ is called the **multiplier** of x_0 . Suppose x_0 is a periodic point of f of period m , then x_0 is said to be,

- a) **attracting** if $0 < |\lambda_m| < 1$;
- b) **super attracting** if $\lambda_m = 0$;
- c) **repelling** if $|\lambda_m| > 1$, and
- d) **indifferent** if $|\lambda_m| = 1$.

Consider the following illustrative example.

Example

Suppose $\mathbb{F} = \mathbb{P}^1(\mathbb{Q}_p)$ where $p \geq 3$ is a prime and consider the mapping $f: \mathbb{P}^1(\mathbb{Q}_p) \mapsto \mathbb{P}^1(\mathbb{Q}_p)$ defined by

$$f(x) = x^2$$

for all $x \in \mathbb{P}^1(\mathbb{Q}_p)$.

The fixed points of f are: 0, 1, and ∞ .

0 is a super attracting point.

1 is an indifferent point.

∞ is an attracting point.

The Fatou set is the whole space while the Julia set is the empty set.

Let x_0 be an m -periodic point and let $h(x) = f^{[m]}(x)$. If there exists an open ball $U(x_0, R)$ of x_0 such that for every $x \in U(x_0, R)$ the following holds

$$\lim_{n \rightarrow \infty} h^{[n]}(x) = x_0$$

then the point x_0 is called an **attractor**. In this event, the set defined by

$$A(x_0) = \left\{ x \in \mathbb{P}^1(\mathbb{K}) : \lim_{n \rightarrow \infty} h^{[n]}(x) = x_0 \right\}$$

is called the **basin of attraction** for x_0 .

Let x_0 be a m -periodic point. If there exists an open ball $U(x_0, \rho)$ ($\rho \in |\mathbb{K}^*|$) such that for every $\rho' < \rho$ ($\rho' \in |\mathbb{K}^*|$), the spheres $S(x_0, \rho')$ are invariant under the mapping $h(x) = f^{[q]}(x)$, then $U(x_0, \rho')$ is said to be a Siegel disk and x_0 is then called a center of a **Siegel disk**. The union of all Siegel disks centered at x_0 is called the **maximal Siegel disk**, which is denoted by $\text{Sl}(x_0)$.

Let $p \geq 2$ **be a prime** and let $N \geq 1$ **be an integer** and let $\mathbb{F} = \mathbb{P}^1(\mathbb{Q}_p)$. Our main goal in this talk consists of studying **the discrete p-adic dynamical systems for the Sigmoid-Beverton-Holt model given by the iterations**,

$$z_{n+1} = f_N(z_n) \tag{1}$$

where $f_N : \mathbb{P}^1(\mathbb{Q}_p) \mapsto \mathbb{P}^1(\mathbb{Q}_p)$ (the **recruitment function** of the Sigmoid Beverton-Holt model) is defined by

$$f_N(z) = \frac{az^N}{1 + z^N}$$

where $a \in \mathbb{Z}_p \setminus \{0\}$ is a constant.

Let us start off by making the following remarks:

- The function f_N **is well-defined** as $f_N(\mathbb{P}^1(\mathbb{Q}_p)) \subset \mathbb{P}^1(\mathbb{Q}_p)$ for all integers $N \geq 1$.
- In the classical context, z_n appearing in Eq. (1) represents **the population size at the generation n** .
- The case $N = 1$ corresponds to the so-called **Pielou logistic model**, which, can be transformed into a **Beverton-Holt model** which arises in fishery.

- The first major challenge in studying the dynamic of Eq. (1) consists of determining **the fixed points of f_N in $\mathbb{P}^1(\mathbb{Q}_p)$** . That is, all the points in $z \in \mathbb{P}^1(\mathbb{Q}_p)$ such that

$$f_N(z) = z.$$

- Clearly, **neither ∞ nor a pole of f_N is a fixed-point for f_N** . Therefore, we have to find all $z \in \mathbb{Q}_p \setminus V_{p,N}$ ($V_{p,N}$ **being the set of all poles, in $\mathbb{Q}_p$, of f_N**) such that

$$z(z^N - az^{N-1} + 1) = 0 \quad (2)$$

which yields f_N has $z = 0$ as a **trivial fixed-point**.

- The main challenge goes back to finding **the other nonzero fixed points**, that is, those $z \in \mathbb{Q}_p \setminus \{V_{p,N} \cup \{0\}\}$ that satisfy

$$z^N - az^{N-1} + 1 = 0. \quad (3)$$

- **Finding solutions to Eq. (3) is one of the most important questions in researching the long-term behavior of the discrete p-adic dynamical systems in Eq. (1).**

- In some cases, we will search for solutions to Eq. (3) in **the finite field** $\mathbb{F}_p = \mathbb{Z}/p\mathbb{Z}$ and then lift those to $\mathbb{Z}_p$ by using the so-called **Hensel Lifting Lemma**.
- However, **this is highly dependent on whether** $|a|_p < 1$ **or** $|a|_p = 1$.
- We shall also see that **the case** $|a|_p = 1$ **is significantly more difficult**.
- **This work builds upon recent progress made in studying non-archimedean rational dynamical systems on the projective line** $\mathbb{P}^1(\mathbb{Q}_p)$.

Consider the function $f_N : \mathbb{P}^1(\mathbb{Q}_p) \mapsto \mathbb{P}^1(\mathbb{Q}_p)$,

$$f_N(z) = \frac{az^N}{1 + z^N}$$

where we have the usual convention $f_N(\infty) = a$, $f_N(\xi) = \infty$ if and only if ξ is a root of the denominator of f_N .

We will be using the following notations in the sequel:

$$\mathbb{Z}_p^* = \{z \in \mathbb{Q}_p : |z|_p = 1\}, \quad V_{p,N} = \{z \in \mathbb{Q}_p : z^N = -1\}, \text{ and}$$

$$U_{p,N} = \{z \in \mathbb{Q}_p : z^N = 1\}$$

where $p \geq 3$ is a prime and $N \geq 1$ is an integer.

Our first goal consists of characterizing the poles of f_N , that is, all $x \neq \infty$ such that

$$x^N + 1 = 0 \text{ in } \mathbb{Q}_p.$$

From $V_{p,N} \subset \mathbb{Z}_p^*$, it follows that it is enough to study the solutions to $x^N + 1 = 0$ in $\mathbb{Z}_p^*$. Now $x^N + 1 = 0$ in $\mathbb{Z}_p^*$ yields $x^N + 1 \equiv 0 \pmod{p}$, which, in turn, is equivalent to the equation,

$$x^N + 1 = 0 \text{ in } \mathbb{F}_p. \tag{4}$$

Let us make the following remarks:

Remark

Note that if N is odd, then $U_{p,N} \subset -V_{p,N}$ where $-V_{p,N} = \{-z : z \in V_{p,N}\}$. Similarly, if N is even, then $V_{p,N} \subset -V_{p,N}$.

In addition to the above remarks, we have the following characterization of the poles of f_N :

Proposition

Suppose $p \geq 3$ is a prime such that $\gcd(p, N) = 1$. Then Eq. (4) has solutions if and only if $\gcd(p-1, N)$ divides $\frac{p-1}{2}$. In this case, $x^N + 1 = 0$ in $\mathbb{Z}_p^$, has $\gcd(p-1, N)$ solutions.*

Proof. Ideas of the proof.

- Recall that $\mathbb{F}_p^*$, the group of units of $\mathbb{F}_p$, is cyclic of order $p-1$ and thus let g denote a generator of it.
- Solving Eq. (4) is equivalent to solving $x^N = g^{\frac{p-1}{2}}$.
- Letting $x = g^y$ for the unknown y it follows that $x^N = g^{Ny} = g^{\frac{p-1}{2}}$ from

which we obtain,

$$Ny \equiv \frac{p-1}{2} \pmod{(p-1)}. \quad (5)$$

- But this is possible if and only if $\gcd(p-1, N)$ divides $\frac{p-1}{2}$.
- In this case, Eq. (5) has $\gcd(p-1, N)$ solutions in $\mathbb{F}_p$.
- It should be mentioned that these $\gcd(p-1, N)$ solutions in $\mathbb{F}_p$ can be computed by solving Eq. (5) which is of the form

$$AX \equiv B \pmod{M}.$$

- To complete the proof, we have to prove that each of these $\gcd(p-1, N)$ solutions to Eq. (5) is also a solution of such an equation in $\mathbb{Z}_p$ (hence in $\mathbb{Q}_p$).
- Indeed, if $\alpha_0 \in \mathbb{F}_p$ is given such that $g_N(\alpha_0) := \alpha_0^N + 1 \equiv 0 \pmod{p}$, then $g'_N(\alpha_0) = N\alpha_0^{N-1}$.
- Since $\gcd(p, N) = 1$, then $g'_N(\alpha_0) = N\alpha_0^{N-1} \not\equiv 0 \pmod{p}$.
- Therefore, by using **Hensel Lifting Lemma**, it follows that there exists a unique $\alpha \in \mathbb{Z}_p$ such that $\alpha^N + 1 = 0$. Obviously, $\alpha \in \mathbb{Z}_p^*$.

We are interested in computing all the $x \in \mathbb{Q}_p \setminus \{V_{p,N} \cup \{0\}\}$ satisfying Eq. (3). In the sequel, we will set

$$h_N(x) := x^N - ax^{N-1} + 1$$

for all $x \in \mathbb{Q}_p \setminus \{V_{p,N} \cup \{0\}\}$.

We will need the following Lemma in the sequel:

Lemma

If $\xi \in \mathbb{Q}_p \setminus \{V_{p,N} \cup \{0\}\}$ is solution to Eq. (3), then $|\xi|_p = 1$.

We have

Proposition

Suppose $p \geq 3$ is a prime such that $\gcd(p, N) = 1$. If $|a|_p < 1$, then Eq. (3) has solutions if and only if $\gcd(p-1, N)$ divides $\frac{p-1}{2}$. In this case, Eq. (3) has $\gcd(p-1, N)$ solutions in $\mathbb{Z}_p^$.*

Proof. If $|a|_p < 1$, then reducing h_N modulo p it follows that $h_N(x) \equiv x^N + 1 \pmod{p}$. Consequently, solving Eq. (3) is reduced to

solving Eq. (4). The rest of the proof is similar to that of the previous Proposition and hence is omitted

Corollary

Suppose $p \geq 3$ is a prime such that $\gcd(p, N) = 1$. If $|a|_p < 1$ and that $\gcd(p-1, N)$ divides $\frac{p-1}{2}$, then the function f_N has $1 + \gcd(p-1, N)$ fixed-points in $\mathbb{P}^1(\mathbb{Q}_p)$ given by

$$\text{Fix}(f_N) = \{0\} \cup \{\omega_{j,N} : j = 1, 2, \dots, \gcd(p-1, N)\}$$

where $\omega_{j,N}$ for $j = 1, 2, \dots, \gcd(p-1, N)$ denote the $\gcd(p-1, N)$ solutions in $\mathbb{Z}_p^$ to Eq. (3). And while 0 is super attractive if $N \geq 2$ and attractive if $N = 1$, all the other fixed-points $\omega_{j,N}$ for $j = 1, 2, \dots, \gcd(p-1, N)$, are repelling.*

Proof.

- First of all, note that the fixed-point 0 is either super attractive or

attractive depending on N . Indeed,

$$f'_N(x) = \frac{aNx^{N-1}}{(1+x^N)^2}$$

for all $x \in \mathbb{P}^1(\mathbb{Q}_p)$, which clearly shows that $|f'_N(0)|_p = 0 < 1$ if $N \geq 2$ and $|f'_N(0)|_p = |aN|_p = |a|_p < 1$ if $N = 1$.

- Using the fact

$$\omega_{j,N}^N - a\omega_{j,N}^{N-1} + 1 = 0$$

for $j = 1, 2, \dots, \gcd(p-1, N)$, we deduce that

$$\left| f'_N(\omega_{j,N}) \right|_p = \left| \frac{aN\omega_{j,N}^{N-1}}{(1+\omega_{j,N}^N)^2} \right|_p = \left| \frac{N}{1+\omega_{j,N}^N} \right|_p = \frac{1}{|a|_p |\omega_{j,N}|_p^{N-1}} = \frac{1}{|a|_p} > 1$$

for $j = 1, 2, \dots, \gcd(p-1, N)$, which clearly shows that all the $\omega_{j,N}$ for $j = 1, 2, \dots, \gcd(p-1, N)$, are repelling.

Consider the following example:

Example

Consider the case $N = 4$, $p = 17$, and pick $a \in \mathbb{Z}_p$ such $|a|_{17} < 1$. Obviously, all the assumptions of Corollary 19 are fulfilled. Therefore, the function f_4 has 4 repelling fixed points in $\mathbb{Z}_{17}$ which are explicitly given by $\omega_{1,4}$, $\omega_{2,4}$, $\omega_{3,4}$, and $\omega_{4,4}$, where

$$\omega_{1,4} \equiv 2 \pmod{17},$$

$$\omega_{2,4} \equiv 8 \pmod{17},$$

$$\omega_{3,4} \equiv 9 \pmod{17},$$

and

$$\omega_{4,4} \equiv 15 \pmod{17}.$$

Consider the case $N = 1$ which corresponds to the Beverton-Holt model which plays a crucial role in population dynamics.

The recruitment function for the Beverton-Holt model, which corresponds to f_1 in this setting, is given by

$$f_1(x) = \frac{ax}{1+x}$$

where $a \in \mathbb{Z}_p \setminus \{0\}$ and $x \in \mathbb{P}^1(\mathbb{Q}_p)$.

The only pole of f_1 is given by $V_{p,1} = \{-1\}$. Computing the first derivative of f_1 , one obtains

$$f_1'(x) = \frac{a}{(1+x)^2}.$$

Theorem

If $a = 1$, then the dynamical system $(\mathbb{P}^1(\mathbb{Q}_p), f_1)$ has an indifferent fixed point given by $x = 0$. Moreover, the maximal Siegel disc around such a fixed point is given by $Sl(0) = U(0, 1)$.

Proof. Since $a = 1$, then $Fix(f_1) = \{0\}$. Moreover, $|f_1'(0)|_p = 1$ and thus the fixed-point 0 is indifferent. Moreover,

$$f_1^{[n]}(x) = \frac{x}{1 + nx}$$

for all $x \in \mathbb{P}^1(\mathbb{Q}_p)$. Furthermore, for $x \in S(0, r)$ where $r > 0$, $f_1^{[n]}(x) \subseteq S(0, r)$ for $r < 1$. This clearly shows that the maximal Siegel disc around the fixed-point $x = 0$ is $Sl(0) = U(0, 1)$.

We have $\mathcal{J}_{f_1} = \emptyset$ and $\mathcal{F}_{f_1} = \mathbb{P}^1(\mathbb{Q}_p)$. Moreover, the only periodic orbit of f_1 is that of the fixed-point 0, with $O_{f_1}(0) = \{0\}$.

Theorem

If $a \neq 1$, then the dynamical system $(\mathbb{P}^1(\mathbb{Q}_p), f_1)$ has two distinct fixed points, $x_1 = 0$ and $x_2 = a - 1$. Moreover, the following hold:

- The fixed point 0 is attractive if $|a|_p < 1$ and its corresponding basin of attraction is $A(0) = \mathbb{P}^1(\mathbb{Q}_p) \setminus \{a - 1\}$.
- The fixed point 0 is indifferent if $|a|_p = 1$ and its maximal Siegel disc is $Sl(0) = U(0, 1)$.
- The fixed point $a - 1$ is repelling if $|a|_p < 1$.
- The fixed point $a - 1$ is indifferent if $|a|_p = 1$ and its corresponding maximal Siegel disc is $Sl(a - 1) = U(a - 1, 1)$.

Proof. Ideas of the proof.

- Since $a \neq 1$, the function f_1 has two distinct fixed points, $x_1 = 0$ and $x_2 = a - 1$.
- Now $|f'_1(0)|_p = |a|_p \leq 1$ and $|f'_1(a - 1)|_p = \frac{1}{|a|_p} \geq 1$. Therefore, the fixed point 0 is attractive if $|a|_p < 1$ and indifferent if $|a|_p = 1$.

- Similarly, the fixed point $a - 1$ is repelling if $|a|_p < 1$ and indifferent if $|a|_p = 1$.
- Using induction, it can be easily seen that

$$f_1^{[n]}(x) = \frac{a^n x}{s_n(a)x + 1}$$

for all $x \in \mathbb{P}^1(\mathbb{Q}_p)$ where $s_n(a) := \frac{1 - a^n}{1 - a}$ for $n \in \mathbb{N}$.

a) Suppose $|a|_p < 1$. Thus $|s_n(a)|_p = 1$ for all $n \in \mathbb{N}$.

If $|x|_p > 1$, then $|f^{[n]}(x)|_p = |a|_p^n \rightarrow 0$ as $n \rightarrow \infty$. Similarly, if $|x|_p < 1$, then $|f^{[n]}(x)|_p = |a|_p^n |x|_p \rightarrow 0$ as $n \rightarrow \infty$. If $x \in \mathbb{Z}_p^* \setminus \{a - 1\}$, it is also clear that $|f^{[n]}(x)|_p \rightarrow 0$ as $n \rightarrow \infty$. Now at the point $x = a - 1$, we have $\lim_{n \rightarrow \infty} |s_n(a)x + 1|_p = 0$, which yields $|f^{[n]}(x)|_p \rightarrow \infty$ as $n \rightarrow \infty$.

Therefore,

$$A(0) = \mathbb{P}^1(\mathbb{Q}_p) \setminus \{a - 1\}.$$

b) Suppose $|a|_p = 1$. In this case both $x_1 = 0$ and $x_2 = a - 1$ are indifferent.

Now for $x \in S(0, r)$ where $r > 0$, $f_1^{[n]}(x) \subseteq S(0, r)$ for $r < 1$. This clearly shows that the maximal Siegel disc around the fixed-point $x = 0$ is $\text{Sl}(0) = U(0, 1)$.

Similarly, taking $x \in S(a - 1, r)$ where $r > 0$ it follows that $f_1^{[n]}(x) \subseteq S(a - 1, r)$ for $r < 1$, which yields the maximal Siegel disc associated with the fixed-point $x = a - 1$ is

$$\text{Sl}(0) = U(a - 1, 1).$$

- If $|a|_p < 1$, one has $\mathcal{J}_{f_1} = \{a - 1\}$ and $\mathcal{F}_{f_1} = \mathbb{P}^1(\mathbb{Q}_p) \setminus \{a - 1\}$.
- If $a = -1$, then $f_1(-1) = \infty$, $f_1^{[2]}(-1) = f_1(\infty) = -1$, $f_1^{[3]}(-1) = \infty$, etc.
- This means both ∞ and -1 are 2-periodic points and that $O_{f_1}(\infty) = \{\infty, -1\} = O_{f_1}(-1)$.
- Now, $(f_1^{[2]})'(x) = 1$, which means both -1 and ∞ are indifferent periodic points. In this case, $\mathcal{J}_{f_1} = \emptyset$ and $\mathcal{F}_{f_1} = \mathbb{P}^1(\mathbb{Q}_p)$.

Suppose $N \geq 2$ and consider the Sigmoid-Beverton-Holt model whose recruitment function is given by $f_N : \mathbb{P}^1(\mathbb{Q}_p) \mapsto \mathbb{P}^1(\mathbb{Q}_p)$,

$$f_N(x) = \frac{ax^N}{1+x^N}$$

where $a \in \mathbb{Z}_p \setminus \{0\}$.

THE CASE $|a|_p < 1$.

Theorem

Suppose $p \geq 3$ is a prime and that $\gcd(p, N) = 1$. If $|a|_p < 1$ and that $\gcd(p-1, N)$ divides $\frac{p-1}{2}$, then the dynamical system $(\mathbb{P}^1(\mathbb{Q}_p), f_N)$ has:

- ① *An attractive point $x = 0$. Moreover, the basin of attraction associated with such a fixed point is given by $A(0) = U(0, 1)$.*
- ② *The other $\gcd(p-1, N)$ fixed-points, that is, $\omega_{j,N}$ for $j = 1, 2, \dots, \gcd(p-1, N)$, are repelling.*

We have the following remark:

Remark

The corresponding Julia and Fatou's sets are respectively given by

$$\mathcal{J}_{f_N} = \{\omega_{j,N}, j = 1, 2, \dots, \gcd(p-1, N)\},$$

$$\mathcal{F}_{f_N} = \mathbb{P}^1(\mathbb{Q}_p) \setminus \{\omega_{j,N}, j = 1, 2, \dots, \gcd(p-1, N)\}.$$

THE CASE $|a|_p = 1$.

Letting $|a|_p = 1$ means that

$$a \equiv q \pmod{p} \text{ where } q = 1, 2, \dots, p-1.$$

Reducing the polynomial h_N modulo p , it follows that

$$h_N(z) = z^N - qz^{N-1} + 1 \in \mathbb{F}_p[z].$$

Consequently, locating the nonzero fixed-point of f_N in the special case $|a|_p = 1$ amounts to finding the roots of

$$h_N(z) = z^N - qz^{N-1} + 1 \in \mathbb{F}_p[z]$$

where $q = 1, 2, \dots, p-1$.

This problem seems to be more challenging and hence will be left as an open problem. However, we have the following partial result.

Proposition

Suppose $N = 2$ and let $p \geq 5$ be a prime such that $3 \mid p-1$. If $a \equiv 1 \pmod{p}$ or $a \equiv -1 \pmod{p}$, then f_2 has three fixed points.

Proof

a) Suppose $a \equiv 1 \pmod{p}$.

- Consequently, $h_2(z) = z^2 - z + 1 \in \mathbb{F}_p[z]$.
- Similarly, $h_2(-z) = z^2 + z + 1 \in \mathbb{F}_p[z]$ whose roots belong to $\overline{\mathbb{F}_p}^{alg}$, are the roots of, $z^3 = 1$ in $\mathbb{F}_p$, whenever $p \neq 3$.
- Such a root belongs to $\mathbb{F}_p$ if and only if $3 \mid p - 1$. In this event, let $\delta \in \mathbb{F}_p$ such that $\delta^3 = 1$ and $\delta \neq 1$.
- Therefore, $h_2(-\delta) = 0$ in $\mathbb{F}_p$. Moreover, $h'(-\delta) = 2\delta - 1 \neq 0$ in $\mathbb{F}_p$.
- Applying the Hensel Lifting Lemma it follows that there exists $\gamma \in \mathbb{Z}_p$ such that $h_2(\gamma) = 0$ with $\gamma \equiv -\delta \pmod{p}$.
- Consequently, the fixed-points of f_2 are 0, γ , and $-\gamma$.

b) Suppose $a \equiv -1 \pmod{p}$.

- Consequently, $h_2(z) = z^2 + z + 1 \in \mathbb{F}_p[z]$.
- If $3 \mid p - 1$, then there exists $\sigma \in \mathbb{F}_p$ such that $\sigma^3 = 1$ in $\mathbb{F}_p$ with $h(\sigma) = 0$ and $h'_2(\sigma) = 2\sigma + 1 \neq 0$ in $\mathbb{F}_p$.
- Applying the Hensel Lifting Lemma it follows that there exists $\xi \in \mathbb{Z}_p$ such that $h_2(\xi) = 0$ with $\xi \equiv \sigma \pmod{p}$.
- Therefore, the fixed-points of f_2 are 0, ξ , and $-\xi$.

- Let us take a closer look at the three fixed points of the function f_2 under the assumptions of the previous Proposition.
- Indeed, note that $\gcd(p-1, 2) = 2$.
- Moreover,

$$\left| f_2'(\omega_{j,2}) \right|_p = \frac{1}{|a|_p |\omega_{j,2}|_p^2} = 1$$

for $j = 1, 2$, which clearly shows that $\omega_{j,2}$ for $j = 1, 2$ (take $\omega_{1,2} = \gamma$ and $\omega_{2,2} = -\gamma$ in the case when $a \equiv 1 \pmod{p}$ and $\omega_{1,2} = \xi$ and $\omega_{2,2} = -\xi$ in the case when $a \equiv -1 \pmod{p}$), are indifferent.

Let us make a few remarks on the Fatou and Julia sets associated with the discrete p-adic dynamical system $(\mathbb{P}^1(\mathbb{Q}_p), f_N)$ in the special case $|a|_p = 1$.

Indeed, although it appears difficult to determine the nonzero fixed points of f_N in this case, using **the method of reduction of rational functions on $\mathbb{P}^1(\mathbb{Q}_p)$** , it can be seen that the sequence of iterates $\{f_N^{[n]}\}_{n \geq 0}$ is equicontinuous on $\mathbb{P}^1(\mathbb{Q}_p)$. Therefore,

$$\mathcal{F}_{f_N} = \mathbb{P}^1(\mathbb{Q}_p) \text{ and } \mathcal{J}_{f_N} = \emptyset.$$

We have compiled a list of research-worthy questions that remain unanswered on this subject.

- Compute all the fixed points of f_N in the special case $|a|_p = 1$ and study the long-term behavior of the dynamical system $(\mathbb{P}^1(\mathbb{Q}_p), f_N)$ around them.
- Study all the periodic orbits of the dynamical system $(\mathbb{P}^1(\mathbb{Q}_p), f_N)$.

- ① S. Albeverio, P. E. Kloeden, and A. Khrennikov, Human memory as a p-adic dynamic system. *Theor Math Phys* **117**, 1414-1422 (1998).
- ② S. Albeverio, A. Khrennikov, B. Tirozzi, and S. De Smedt, p-adic dynamic systems. *Theor Math Phys* **114** (1998), pp. 276-287.
- ③ V. Anashin and A. Khrennikov, Applied algebraic dynamics. De Gruyter Expositions in Mathematics, 49. Walter de Gruyter & Co., Berlin, 2000.
- ④ R. L. Benedetto, *Dynamics in one non-archimedean variable*. Graduate Studies in Mathematics, 198. American Mathematical Society, Providence, RI, 2019.
- ⑤ T. Bezandry, T. Diagana, and S. Elaydi, On the stochastic Beverton-Holt equation with survival rates. *J. Diff. Equations Appl.*, **14** (2008), pp. 175-190.

- ⑥ W. Bradford, S. Lake, T. Tappin, and S. Weatherby, *Stability in the Sigmoid-Beverton-Holt model*, 2010 SMILE Program, Xavier University of Louisiana, Available on: math.lsu.edu/vigre/2010SMILE 2010.
- ⑦ G. Call, J. Silverman, Canonical height on varieties with morphisms, *Compos. Math.* **89** (1993), pp. 163–205.
- ⑧ T. Diagana, The Long-term Behavior of the p-adic Sigmoid Beverton-Holt Dynamical Systems in the Projective Line $\mathbb{P}^1(\mathbb{Q}_p)$. *Journal of Mathematical Analysis and Applications*. Vol. **530**, Issue 1, 1 February 2024, 127638.
- ⑨ T. Diagana and F. Ramaroson, *Non-archimedean operator theory*. Springer, New York, 2016. 156 pp.
- ⑩ T. Diagana and H. Maiga, Some new identities and congruences for Fubini numbers. *J. Number Theory* **173** (2017), pp. 547-569.

- 11 T. Diagana, S. Elaydi, and A. A. Yakubu, Population models in almost periodic environments. *J. Diff. Equations Appl.*, **13** (2007), no. 4, pp. 239-260.
- 12 B. Diarra and D. Sylla, p -adic dynamical systems of Chebyshev polynomials. *p-Adic Numbers Ultrametric Anal. Appl.*, **6** (1) (2014), pp. 21-32.
- 13 B. Diarra, The Lipschitz condition for rational functions on ultrametric valued fields. *Advances in non-Archimedean analysis*, 91–103, Contemp. Math., 551, Amer. Math. Soc., Providence, RI, 2011.
- 14 A. Fan, S. Fan, L. Liao, and Y. Wang, On minimal decomposition of p -adic homographic dynamical systems. *Adv. Math.* **257** (2011), no. 1, pp. 92-135.
- 15 A. Fan and L. Liao, On minimal decomposition of p -adic polynomial dynamical systems. *Adv. Math.* **228** (2011), no. 4, pp. 2116-2144.

- 16 A. Fan and L. Liao, Dynamical structures of Chebyshev polynomials on $\mathbb{Z}_2$. *J. Number Theory* **169** (2016), pp. 174-182.
- 17 G. R.J. Gauta, K. Goldringa, F. Grogana, C. Haskellb and R. J. Sacker, Difference equations with the Allee effect and the periodic Sigmoid-Beverton-Holt equation revisited. *J. Biol. Dynamics* **6** (2012), no. 2, pp. 1019-1033.
- 18 M. Gundlach, A. Khrennikov, and K. Lindahl, On ergodic behavior of p-adic dynamical dystems. *Infinite Dimensional Analysis, Quantum Probability and Related Topics* 4 (2001), pp. 569–577.
- 19 A. Harry, C. M. Kent, and V. L. Kocic, *Global behavior of solutions of a periodically forced Sigmoid-Beverton-Holt model*, *J. Biol. Dynamics*, **6** (2012), no. 2, pp. 212-234.
- 20 C. M. Kent, V. L. Kocic, and Y. Kostrov, Attenuance and resonance in a periodically forced Sigmoid-Beverton-Holt model. *Int. J. Difference Equ.* **7** (2012), no. 1, 35-60.

- 21 A. Khrennikov, *Non-archimedean analysis: quantum paradoxes, dynamical systems and biological models*. Mathematics and its Applications, 427. Kluwer Academic Publishers, Dordrecht, 1997.
- 22 A. Khrennikov, *p -adic valued distributions in mathematical physics*. Mathematics and its Applications, 309. Kluwer Academic Publishers Group, Dordrecht, 1994.
- 23 A. Khrennikov, p -adic quantum mechanics with p -adic valued functions. *J. Math. Phys.* **32** (1991), no. 4, pp. 932-937.
- 24 A. Khrennikov and M. Nilsson, On the number of cycles of p -adic dynamical systems. *J. Number Theory* **90** (2001), no. 2, pp. 255–264.
- 25 A. Y. Khrennikov and M. Nilson, *p -adic deterministic and random dynamics*. Mathematics and its Applications, **574**. Kluwer Academic Publishers, Dordrecht, 2004.
- 26 J. Knight, *Dynamical systems of p -adic $(3, 3)$ -rational functions*. Thesis (Ph.D.)-University of Alabama in Huntsville. 2022. 440 pp.

- 27 F. Mukhamedov, On the chaotic behavior of cubic p-adic dynamical systems. *Mathematical Notes* **83** (2008), pp. 428-431.
- 28 K. Mukhamedov and Wan N. F. A. W. Rozali, On a p-adic Cubic Generalized Logistic Dynamical System. *Journal of Physics Conference Series* **435** (2013), no. 1, Id: 012012.
- 29 F. Mukhamedov and J. Mendes, On the chaotic behaviour of a generalized Logistic p-adic dynamical system. *J. Differential Equations* **243** (2007), pp. 125-145.
- 30 F. Mukhamedov and U. Rozikov, A polynomial p-adic dynamical system. *Theor Math Phys* **170** (2012), pp. 376-383.
- 31 F. Mukhamedov and M. Saburov, On the equation $x^q = a$ over $\mathbb{Q}_p$. *J. Number Theory* **133** (2013), no. 1, pp. 55-58.
- 32 U. Rozikov and I. Sattarov, Dynamical systems of the p-Adic (2, 2)-rational functions with two fixed points. *Results Math.* **75** (2020), no. 3, Paper No. 100, 37 pp.

- 33 U. Rozikov and I. Sattarov, On a nonlinear p -adic dynamical system. *P-Adic Numbers, Ultrametric Analysis, and Applications* **6** (2013), pp. 54-65.
- 34 U. Rozikov and I. Sattarov, p -adic dynamical systems of $(2,2)$ -rational functions with unique fixed point. *Chaos Solitons & Fractals* **105** (2017), pp. 260-270.
- 35 I.A.Sattarov, p -adic $(3,2)$ -rational dynamical systems with three fixed points. *Uzbek Math. J.* 2019, no. 3, pp. 133-145.
- 36 I. Sattarov, p -Adic $(3, 2)$ -rational dynamical systems. *P-Adic Numbers, Ultrametric Analysis, and Applications* **7** (2015), pp. 39-55.
- 37 W. Schikhof, *Ultrametric calculus*. Cambridge Univ. Press, Cambridge (1984).
- 38 J. Silverman, *The arithmetic of dynamical systems*. Graduate Texts in Mathematics, no. 241, Springer-Verlag, New York, 2007.

- 39 I. Volovich, Number theory as the ultimate physical theory. *P-Adic Numbers, Ultrametric Analysis, and Applications* **2** (1987), pp. 77-87.

THANK YOU — *p*-adically

ANY QUESTIONS?